

OAEP (Optimal Asymmetric Encryption Padding)

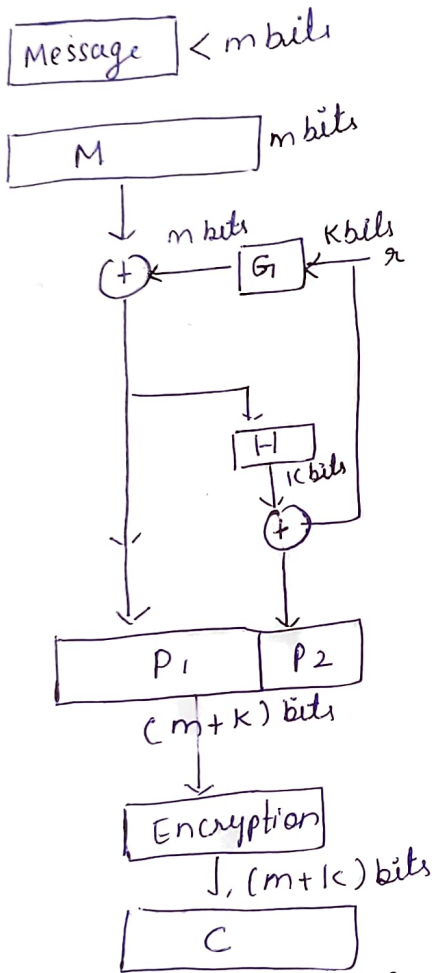
Encryption Padding

P : Plaintext ($P_1 || P_2$) G : public function
(k -bit to m -bit)

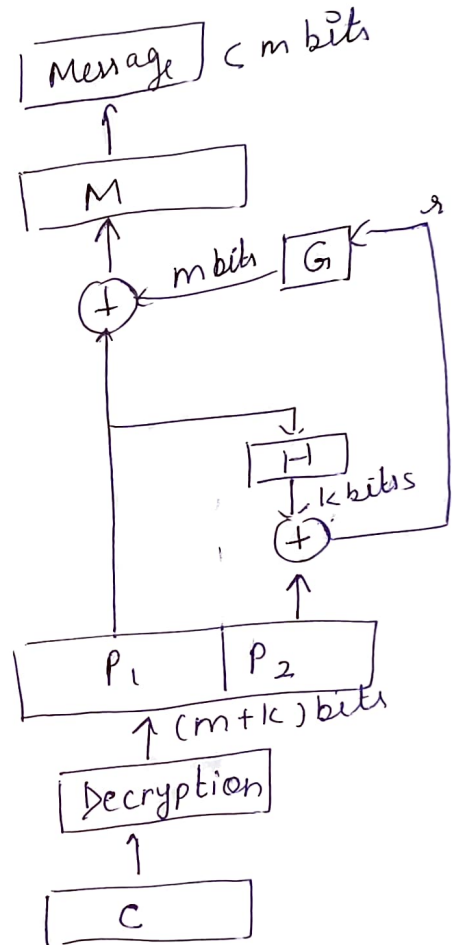
C : Ciphertext

H : public function
(m to k -bits)

M - Padded message
 r - One-time random
number



Sender



Receiver

(IND-CPA
Secure)

OAEP (Optimal Asymmetric Encryption
padding)

1. For message and do padding to size
(indistinguishability under chosen plaintext attack)
 $m \rightarrow M$ (increase size of message)
(padding or masking)

(2) $r \rightarrow k \text{ bits}$ (used only once)
randomly generate r every time

(3) From this r (k -bits) generate m bit
integer using $G(r)$ [Masking or padding]

(4) Apply masking to create first part
of plaintext - $P_1 = M \oplus G(r)$

(5) Generate $P_2 = H(P_1) \oplus r$
(hash function)
function m bit input k -bit output

(6) $P = P_1 \parallel P_2$

(7) $c = P^e \bmod n$

Decryption

(1) $P = c^d \bmod n$

$= P_1 \parallel P_2$

(2) recreate r

$$H(P_1) \oplus P_2 = H(P_1) \oplus H(P_2) \oplus r = r$$

(3) Remove mask $G(r) \oplus P_1$

$$G(r) \oplus G(r) \oplus M = M$$

Problems with RSA (Textbook)

- No randomness $\rightarrow$ Insecure
(use randomness)
- much slower than secret-key crypto
 - use hybrid-encryption $\text{RSA}(K), \text{AES}(K, M)$
- Not secure against chosen ciphertext attack.

USE OAEP which solves all prob above

RSA Encryption + OAEP (CCA secure)
padding scheme

- public key (n, e)
- private key (p, q, d)

Enc

$$C = \text{OAEP}(m, 00 \dots 00, r)^e \bmod n$$

Dec

$$(m, r) = \text{OAEP}^{-1}(C^d \bmod n)$$

if $! = 00 \dots 00$ abort
else o/p m

